



Separation of benzene from alkanes by solvent extraction with 1-ethylpyridinium ethylsulfate ionic liquid

Elena Gómez^a, Irene Domínguez^b, Noelia Calvar^a, Ángeles Domínguez^{b,*}

^a Laboratory of Separation and Reaction Engineering, Departamento de Engenharia Química, Faculdade de Engenharia, Universidade do Porto, 4200-465 Porto, Portugal

^b Departamento de Ingeniería Química, Universidad de Vigo, 36310 Vigo, Spain

ARTICLE INFO

Article history:

Received 30 November 2009

Received in revised form 19 April 2010

Accepted 26 April 2010

Available online 4 May 2010

Keywords:

Hexane

Heptane

Benzene

(Liquid + liquid) equilibrium

[EPy][EtSO₄] ionic liquid

ABSTRACT

The (liquid + liquid) equilibrium (LLE) data for ternary mixtures {alkane + benzene + 1-ethylpyridinium ethylsulfate ([EPy][EtSO₄])} at $T = (283.15 \text{ and } 298.15) \text{ K}$ and atmospheric pressure are presented. The alkanes used were hexane and heptane. The cloud point method was used to determine the binodal curve, and the tie-line compositions were obtained by density measurements. The LLE data obtained were used to calculate distribution coefficients and selectivity values. The consistency of tie-line data was ascertained by applying the Othmer-Tobias and Hand equations. Correlation of the experimental tie-lines was conducted through the use of NRTL equation, which provides good correlation of the experimental data.

The results show that [EPy][EtSO₄] can be used as an alternative solvent in liquid extraction processes for the removal of benzene from its mixtures with alkanes.

© 2010 Elsevier Ltd. All rights reserved.

1. Introduction

Separation of aromatic hydrocarbon from aliphatic hydrocarbon mixtures has traditionally been one of the most challenging tasks in refinery processes [1], since these hydrocarbons have boiling points in a close range and several combinations form azeotropes, and this separation process cannot be carried out in an efficient and economic way by distillation.

The conventional processes for the separation of these aromatic and aliphatic hydrocarbon mixtures are liquid extraction, extractive distillation and azeotropic distillation [1–4]. Typical solvent in processes cited above are organic compounds such as sulfolane [5–12], ethylene glycols [12–14], *N*-methylpyrrolidone [11] among some others [2–4]. Replacement of volatile solvents in the separation of aromatic and aliphatic by non-volatile ionic liquids (ILs) can offer several advantages: less complex processes and a more simple regeneration of the solvent.

One of the general advantages offered by ionic liquids is their tunability, which has led them to be also known as “designer solvents”: different types of cations and anions can be combined, and their respective structures tailored, to modify the properties of the resulting ionic liquids according to the interest in each application [15]. To apply these ionic liquids in the chemical process industry, thermodynamic data need to be available for designing a proper separation process. Nevertheless, in the literature there are a few

references concerning (liquid + liquid) equilibrium (LLE) for aromatic and aliphatic hydrocarbon with ILs [16–25].

In this work, binodal curve data and tie-line data for the ternary mixtures {alkane (1) + benzene (2) + [EPy][EtSO₄] (3)} were determined at $T = (283.15 \text{ and } 298.15) \text{ K}$ and at atmospheric pressure. From the experimental data, the selectivity, S , and solute distribution ratio, β , were calculated to determine the feasibility of the IL for solvent extraction processes. The consistency of the tie-line data was ascertained by applying the Othmer-Tobias [26] and Hand [27] equations. The experimental data were correlated through the NRTL [28] model, and the results were analyzed in terms of deviations between experimental and calculated compositions in both equilibrium phases and between experimental and calculated distribution coefficient.

2. Experimental section

2.1. Chemicals

Hexane, heptane, and benzene used in this work were supplied by Aldrich with mass fraction purity >99.0, 99.5 and 99.8, respectively. They were degassed ultrasonically and dried over molecular sieves type 4 nm supplied by Aldrich, and kept in an inert argon atmosphere.

The ionic liquid was synthesized and purified in our laboratory following the procedure previously published [29]. The ionic liquid was subjected to vacuum ($P = 2 \cdot 10^{-1} \text{ Pa}$) and moderate temperature ($T = 323.15 \text{ K}$) for several days to remove possible traces of

* Corresponding author. Tel.: +34 986 812 422; fax: +34 986 812 382.

E-mail address: admguez@uvigo.es (Á. Domínguez).

solvents and moisture, always prior to its use. The water content was determined using a 787 Karl Fischer Titrino and the ionic liquid showed that mass fraction of water was less than $7 \cdot 10^{-4}$.

To ensure its purity, the structure of the final product was checked by nuclear magnetic resonance (NMR) spectroscopy. Its purity was found to be higher than 0.99 mass fraction. The ionic liquid was kept in bottles under inert atmosphere in a glovebox.

The experimental physical properties of the pure components at $T = 298.15$ K are listed in table 1, and compared with recent literature data [30,31]. No literature data about the density and refractive index of [EPy][EtSO₄] have been found.

2.2. Apparatus and procedure

The density of the pure liquids and mixtures were measured using an Anton Paar DSA-5000 digital vibrating tube densimeter with an uncertainty of $\pm 3 \cdot 10^{-5} \text{ g} \cdot \text{cm}^{-3}$. To measure refractive indices of pure components an automatic refractometer (Abbe-mat-HP, Dr. Kernchen) with an uncertainty in the experimental measurements of $\pm 4 \cdot 10^{-5}$ was used. For the preparation of samples, a Mettler AX-205 Delta Range balance with an uncertainty of $\pm 3 \cdot 10^{-4} \text{ g}$ was employed.

LLE data for the ternary systems {hexane (1), or heptane (1) + benzene (2) + 1-ethylpyridinium ethylsulfate (3)} were determined at $T = (283.15 \text{ and } 298.15) \text{ K}$. The points on the binodal curves were obtained using the “cloud point” method [32], while the tie-line compositions were determined by density measurements.

The binodal curves of the studied systems were determined at $T = (283.15 \text{ and } 298.15) \text{ K}$ and at atmospheric pressure by titrating binary mixtures (3 ml) of known compositions with the third component until the transition was visually determined. The density of each sample was measured and a polynomial expression for the density as function of composition was obtained. Note that the binary mixture {alkane (1) + benzene (2)} is part of binodal curve, since miscibility of the IL in binary mixtures was not detected.

To estimate the error of the technique used for the determination of the binodal curves, three validation points were used. These points were obtained by weighing, and then their densities were determined. The compositions of these points were calculated through the polynomial expression above mentioned, and the obtained values were compared with the known compositions. The maximum error was estimated to be 0.007 in mole fraction.

For the tie-line determination, mixtures (12 ml) of the compounds with an overall composition lying in the immiscible region were prepared and placed inside glass cells, and closed using a silicon cover, they were thermostatted at $T = (283.15 \text{ and } 298.15) \text{ K}$ in a thermostatic bath (PoliScience digital temperature controller) with a precision of $\pm 0.01 \text{ K}$. Special care was taken in covering the whole immiscibility region, and getting well-distributed tie-lines throughout it. To guarantee the thermodynamic equilibrium, the mixtures were agitated using a magnetic stirrer for 6 h in order

to allow an intimate contact between phases, and then the equilibrium phases were left overnight at the studied temperatures to settle down. The indicated times were determined according to results from preliminary studies. Then, a sample from each phase was withdrawn using a syringe to carry out the compositional analysis. The determination of the tie-line compositions for the ternary systems was carried out by correlating the densities of the two immiscible liquid phases of the conjugate solutions with the curve of density versus composition. No presence of ionic liquid was observed with NMR technique in the upper equilibrium phase of the tie-lines of the studied systems, therefore the composition of this phase was determined using the density curves of the binary systems {alkane (1) + aromatic (2)}, taken from the literature [33].

3. Results and discussion

The composition of the “cloud point” on the binodal curves at $T = (283.15 \text{ and } 298.15) \text{ K}$ for both systems in study are given in table 2, and the validation points are summarized in table 3.

In table 4 the polynomial expressions for the density as function of composition for each binodal curve are presented.

For the study of the ability of the ionic liquid [EPy][EtSO₄] to separate (aliphatic + benzene), (liquid + liquid) equilibrium for the ternary systems {hexane (1) + benzene (2) + 1-ethylpyridinium ethylsulfate (3)} and {heptane (1) + benzene (2) + 1-ethylpyridinium ethylsulfate (3)} were experimentally determined at $T = (283.15 \text{ and } 298.15) \text{ K}$ and at atmospheric pressure. The experimental measured compositions of phases in equilibrium in the ternary mixtures are presented in tables 5 and 6. Figure 1 shows graphical representations of the tie-lines of both ternary systems at the studied temperatures. The feasibility of the ionic liquid [EPy][EtSO₄] as a solvent to perform the extraction of benzene from mixtures with hexane or heptane was evaluated using classic parameters such as the solute distribution ratio, β , and the selectivity, S , calculated from the experimental data. Tables 5 and 6 include these parameters. Their calculation is easily made according to the following expressions:

TABLE 2

Binodal curves of ternary systems {alkane (1) + benzene (2) + [EPy][EtSO₄] (3)}.

x_1	x_2	$\rho/(\text{g} \cdot \text{cm}^{-3})$
{Hexane (1) + benzene (2) + [EPy][EtSO ₄] (3)}		
$T = 283.15 \text{ K}$		
0.007	0.000	1.25982
0.006	0.050	1.25286
0.006	0.099	1.24460
0.004	0.204	1.22558
0.002	0.252	1.21675
0.000	0.335	1.19857
$T = 298.15 \text{ K}$		
0.008	0.000	1.24978
0.007	0.050	1.24250
0.006	0.100	1.23489
0.005	0.202	1.21562
0.003	0.248	1.20663
0.002	0.300	1.19568
0.000	0.406	1.17169
{Heptane (1) + benzene (2) + [EPy][EtSO ₄] (3)}		
$T = 283.15 \text{ K}$		
0.007	0.000	1.26051
0.006	0.051	1.25261
0.003	0.146	1.23638
0.003	0.198	1.22660
0.003	0.243	1.21777
0.001	0.295	1.20712
0.000	0.335	1.19857
$T = 298.15 \text{ K}$		
0.007	0.000	1.25063
0.007	0.048	1.24201
0.005	0.098	1.23387
0.003	0.201	1.21597
0.003	0.251	1.20696
0.001	0.354	1.18460
0.000	0.406	1.17169

TABLE 1

Pure component properties data at $T = 298.15 \text{ K}$.

Component	$\rho/(\text{g} \cdot \text{cm}^{-3})$		n_D	
	Exp.	Lit.	Exp.	Lit.
Hexane	0.65519	0.65484 ^a	1.37234	1.37226 ^a
Heptane	0.67956	0.67946 ^a	1.38515	1.38511 ^a
		0.6786 ^b		
Benzene	0.87357	0.87360 ^a	1.49774	1.49792 ^a
[EPy][EtSO ₄]	1.2520	n.a. ^c	1.50525	n.a. ^c

^a From reference [30].

^b From reference [31].

^c n.a.: not available.

TABLE 3Validation points of ternary systems {alkane (1) + aromatic (2) + [EPy][EtSO₄] (3)}.

x_1	x_2	$\rho/(\text{g} \cdot \text{cm}^{-3})$	x_1^{calc}	x_2^{calc}
{Hexane (1) + benzene (2) + [EPy][EtSO ₄] (3)}				
$T = 283.15 \text{ K}$				
0.006	0.051	1.25243	0.006	0.051
0.004	0.173	1.23216	0.004	0.171
0.002	0.255	1.21455	0.002	0.261
$T = 298.15 \text{ K}$				
0.009	0.053	1.24210	0.007	0.053
0.005	0.173	1.22099	0.007	0.173
0.002	0.292	1.19759	0.002	0.290
{Heptane (1) + benzene (2) + [EPy][EtSO ₄] (3)}				
$T = 283.15 \text{ K}$				
0.003	0.081	1.24799	0.004	0.079
0.003	0.181	1.22984	0.003	0.181
0.003	0.278	1.21065	0.002	0.278
$T = 298.15 \text{ K}$				
0.007	0.075	1.23782	0.006	0.075
0.006	0.178	1.21975	0.004	0.182
0.004	0.263	1.20326	0.002	0.268

$$\beta = x_2^{\text{II}}/x_2^{\text{I}}, \quad (1)$$

$$S = x_2^{\text{II}}x_1^{\text{I}}/x_2^{\text{I}}x_1^{\text{II}}, \quad (2)$$

where x_1^{I} and x_2^{I} are the mole fraction of alkane and benzene, respectively, in the alkane-rich phase, and x_1^{II} and x_2^{II} are the mole fraction of alkane and benzene, respectively, in the IL-rich phase. The evolution of β and S for both systems at the studied temperatures, as a function of the mole fraction of the solute in the organic phase, is plotted in figures 2 and 3, respectively. In these figures can be observed that selectivity decreases with an increase of benzene in alkane-rich phase, as was observed in several published data [17,18,21,34–36]. In figures 4 and 5, experimental and literature selectivities [8,17,19,36–42] for the ternary systems {hexane (1) + benzene (2) + ionic liquid (3)} and {heptane (1) + benzene (2) + ionic liquid (3)} are compared, respectively. As it can be observed in figure 4 for the system {hexane (1) + benzene (2) + ionic liquid (3)}, the selectivity values obtained for the system {hexane (1) + benzene (2) + [EPy][EtSO₄] (3)} are higher than those obtained with other ionic liquids, except for the system containing [Emim][EtSO₄] and are similar to those obtained with [Bmim][MeSO₄]. Figure 5 shows that at lower compositions of benzene in the upper phase, the selectivities obtained for the system studied in this work are similar to the literature selectivities for the system containing [EmPy][EtSO₄], while for higher compositions, S values are better with the ionic liquid studied in this work.

The effect of the temperature is rather small, with the characterising parameters of the extraction (S and β) being slightly higher for the lower temperatures. The S values also increase in the order heptane > hexane, and the solute distribution ratio increases in the order hexane > heptane. Interestingly, a parallelism can be established between the evolution of the solute distribution ratio of aliphatic hydrocarbons in the ternary systems {alkane (1) + aromatic (2) + ionic liquid (3)} and the evolution of the solubilities of aliphatic in the ionic liquid. These results are in agreement with the works presented by Letcher *et al.* [36] and Arce and co-workers [23]. Considering the type of ternary diagrams and that a small variation in the compositions has great effect on selectivity, this parameter should be interpreted as a range. However, the selectivity values in the systems in study are higher than unity, which confirms the extraction efficiency of this ionic liquid. Moreover, the use of [EPy][EtSO₄], and of ionic liquids in general, in solvent extraction is favourable because it can be easily recovered and reused.

The equations provided by Othmer-Tobias [26] and Hand [27] were used to ascertain the reliability of the experimental tie-line compositions:

TABLE 4

Polynomial expressions for the density as function of composition for each binodal curve at every studied temperature for both ternary systems.

{Hexane (1) + benzene (2) + [EPy][EtSO ₄] (3)} at $T = 283.15 \text{ K}$	
$\rho/(\text{g} \cdot \text{cm}^3) = 1.2653 - 0.7774 \cdot x_1 - 0.0061 \cdot x_1^2 - 0.0001 \cdot x_1^3 - 0.1455x_2 - 0.1706 \cdot x_2^2 + 0.0321 \cdot x_2^3$	
{Hexane (1) + benzene (2) + [EPy][EtSO ₄] (3)} at $T = 298.15 \text{ K}$	
$\rho/(\text{g} \cdot \text{cm}^3) = 1.2534 - 0.4564 \cdot x_1 + 0.0171 \cdot x_1^2 + 0.0010 \cdot x_1^3 - 0.1398 \cdot x_2 - 0.2043 \cdot x_2^2 + 0.1288 \cdot x_2^3$	
{Heptane (1) + benzene (2) + [EPy][EtSO ₄] (3)} at $T = 283.15 \text{ K}$	
$\rho/(\text{g} \cdot \text{cm}^3) = 1.2604 + 0.0118 \cdot x_1 + 0.0112x_1^2 + 0.0003 \cdot x_1^3 - 0.1484x_2 - 0.1195 \cdot x_2^2 + 0.0360 \cdot x_2^3$	
{Heptane (1) + benzene (2) + [EPy][EtSO ₄] (3)} at $T = 298.15 \text{ K}$	
$\rho/(\text{g} \cdot \text{cm}^3) = 1.2510 - 0.0547 \cdot x_1 - 0.0056 \cdot x_1^2 + 0.0001 \cdot x_1^3 - 0.1807x_2 - 0.1109 \cdot x_2^2 - 0.3634 \cdot x_2^3$	

TABLE 5Experimental (liquid + liquid) equilibrium data in mole fraction for the ternary system {hexane (1) + benzene (2) + [EPy][EtSO₄] (3)} at several temperatures, solute distribution ratio, β , and selectivity, S , values.

Alkane-rich layer		Ionic liquid-rich layer		β	S
x_1^I	x_2^I	x_1^{II}	x_2^{II}		
$T = 283.15\text{ K}$					
0.889	0.111	0.006	0.061	0.55	81.42
0.816	0.184	0.006	0.094	0.51	69.48
0.748	0.252	0.005	0.122	0.48	72.43
0.686	0.314	0.005	0.147	0.47	64.23
0.618	0.382	0.005	0.173	0.45	55.98
0.546	0.454	0.005	0.198	0.44	47.62
0.470	0.530	0.004	0.220	0.42	48.77
0.404	0.596	0.004	0.240	0.40	40.67
0.332	0.668	0.004	0.263	0.39	32.68
$T = 298.15\text{ K}$					
0.888	0.112	0.007	0.055	0.49	62.30
0.826	0.174	0.007	0.084	0.48	56.97
0.751	0.249	0.007	0.15	0.46	49.55
0.680	0.320	0.006	0.142	0.44	50.29
0.623	0.377	0.006	0.165	0.44	45.44
0.544	0.456	0.006	0.194	0.43	38.57
0.473	0.527	0.006	0.221	0.42	33.06
0.409	0.591	0.005	0.239	0.40	33.08
0.250	0.750	0.004	0.289	0.39	24.88

TABLE 6Experimental (liquid + liquid) equilibrium data in mole fraction for the ternary system {heptane (1) + benzene (2) + [EPy][EtSO₄] (3)} at several temperatures, solute distribution ratio, β , and selectivity, S , values.

Alkane-rich layer		Ionic liquid-rich layer		β	S
x_1^I	x_2^I	x_1^{II}	x_2^{II}		
$T = 283.15\text{ K}$					
0.942	0.058	0.005	0.029	0.50	94.20
0.866	0.134	0.005	0.065	0.49	84.01
0.808	0.192	0.005	0.090	0.47	75.75
0.745	0.255	0.005	0.116	0.45	67.78
0.632	0.368	0.004	0.159	0.43	68.27
0.538	0.462	0.004	0.194	0.42	56.48
0.478	0.522	0.004	0.215	0.41	49.22
0.415	0.585	0.004	0.235	0.40	41.68
0.351	0.649	0.003	0.254	0.39	45.79
$T = 298.15\text{ K}$					
0.941	0.059	0.006	0.028	0.47	73.10
0.882	0.118	0.006	0.054	0.46	67.27
0.830	0.170	0.006	0.077	0.45	62.66
0.774	0.226	0.006	0.100	0.44	57.08
0.689	0.311	0.005	0.134	0.43	59.37
0.633	0.367	0.005	0.155	0.42	53.47
0.544	0.456	0.005	0.189	0.41	45.09
0.421	0.579	0.004	0.232	0.40	42.17
0.346	0.654	0.004	0.258	0.39	34.12

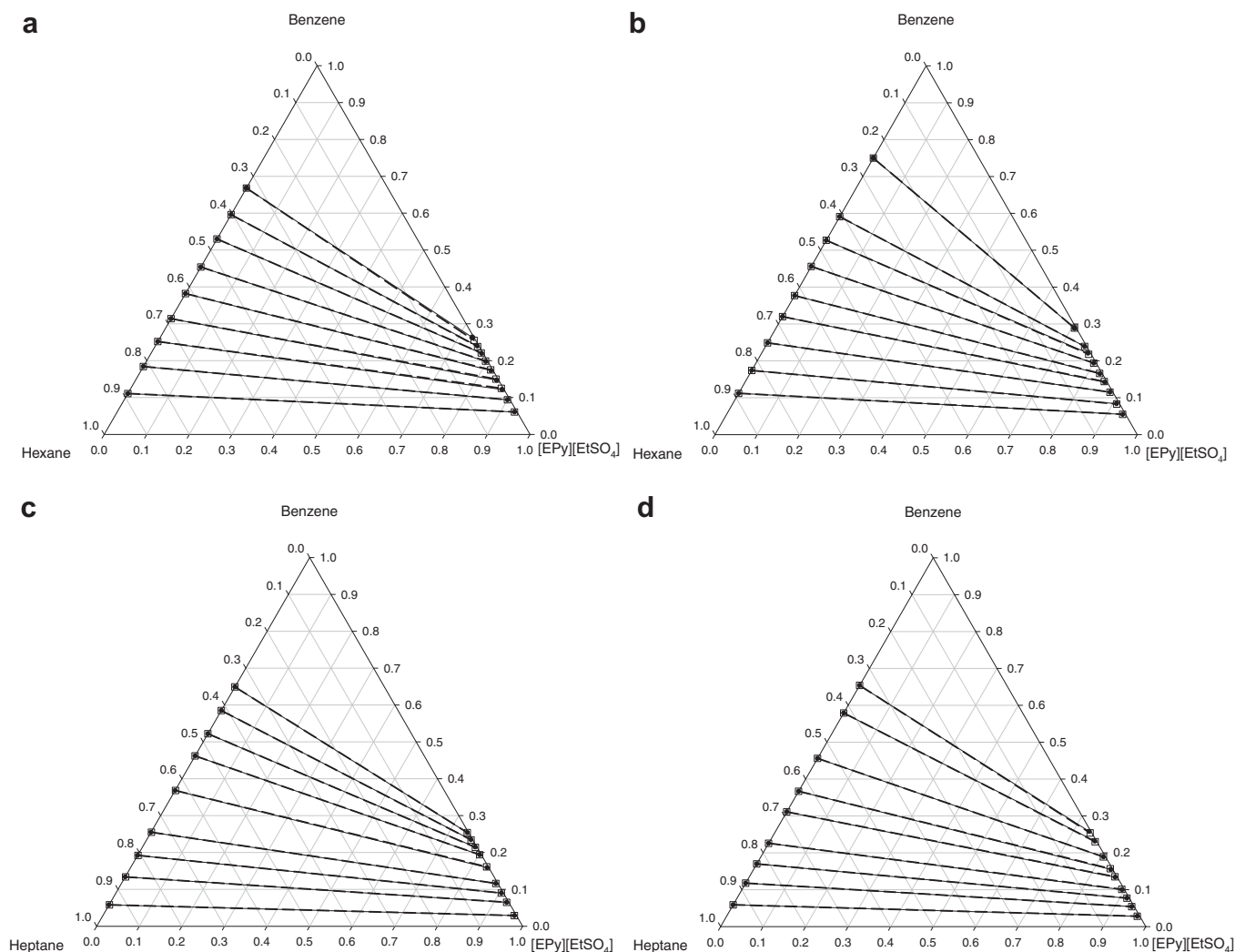


FIGURE 1. Experimental LLE of the ternary systems {hexane (1)+benzene (2)+[EPy][EtSO₄] (3)}: (a) $T=283.15$ K; (b) $T=298.15$ K, and {heptane (1)+benzene (2)+[EPy][EtSO₄] (3)}: (c) $T=283.15$ K; and (d) $T=298.15$ K. Solid lines and full points indicate experimental tie-lines, and dashed lines and empty squares indicate calculated data from NRTL model.

$$\ln(1 - w'_1/w'_1) = a + b \ln(1 - w''_2/w''_2), \quad (3)$$

$$\ln(1 - w'_2/w'_1) = c + d \ln(1 - w''_2/w''_2), \quad (4)$$

where w'_1 and w'_2 are the mass fraction of alkane and benzene, respectively, in the alkane-rich phase; w''_2 and w''_3 are the mass fraction of benzene and ionic liquid, respectively, in the IL-rich phase; and a , b , c , and d are adjustable parameters. The linearity of these fittings indicates the degree of consistency of the experimental data. The parameters obtained from the proposed equations are presented in table 7, together with the correlation factor, R^2 , for both systems at the studied temperatures. As can be inferred from R^2 values presented in this table, the experimental data show a satisfactory linearity.

4. Thermodynamic correlation

The correlation of the LLE data is interesting as it permits a computational treatment. In this case, the NRTL equation [28] was used to correlate the experimental data. In spite of being a model initially developed for non-electrolytes, previous works demonstrate that the original NRTL equation can successfully correlate LLE data of ternary systems involving IL [17,36,38–40,42]. The α values were settled to different values between 0.1 and 0.4, and the best results found for the correlation were with α equal to 0.1 for both systems at $T=(283.15$ and $298.15)$ K. The binary interaction

parameters, Δg_{ij} , are estimated from experimental data. The objective function used minimizes the differences between the experimental and calculated mole fraction of the components in both phases.

The NRTL binary interaction parameters of the ternary systems correlated are listed in table 8, as well as the values of the corresponding deviations. These deviations were calculated as follows:

$$\sigma x = 100 \sqrt{\frac{\sum_i^M \sum_j^{N-1} (x_{ij}^{I,\text{exptl}} - x_{ij}^{I,\text{calc}})^2 + (x_{ij}^{II,\text{exptl}} - x_{ij}^{II,\text{calc}})^2}{2MN}}, \quad (5)$$

$$\Delta \beta = 100 \sqrt{\frac{1}{M} \sum_k \left(\frac{\beta_k - \beta_k^{\text{calc}}}{\beta_k} \right)^2}, \quad (6)$$

where M is the number of tie-lines, and N the number of components of the mixture.

The correlated tie-lines are plotted in figure 1 along with the experimental ones. As expected, the difference between the parameter sets for each temperature is small. The visual comparison of both sets of tie-lines for each system supports the general validity of NRTL correlations carried out.

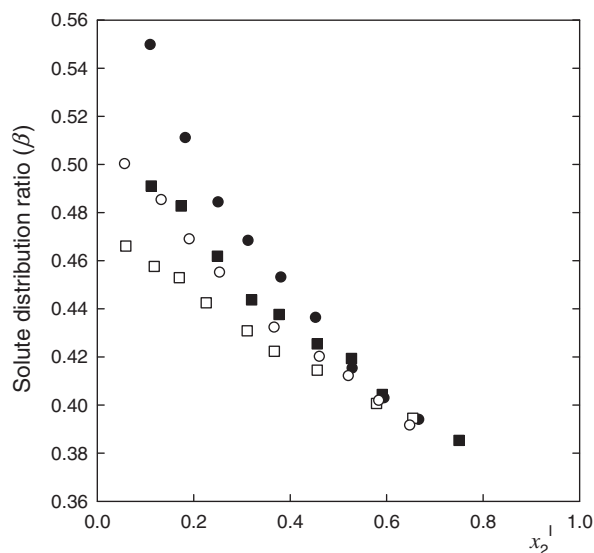


FIGURE 2. Solute distribution ratio for the ternary systems {hexane (1) + benzene (2) + [EPy][EtSO₄] (3)} at $T = 283.15$ K (●), {hexane (1) + benzene (2) + [EPy][EtSO₄] (3)} at $T = 298.15$ K (■), {heptane (1) + benzene (2) + [EPy][EtSO₄] (3)} at $T = 283.15$ K (○), and {heptane (1) + benzene (2) + [EPy][EtSO₄] (3)} at $T = 298.15$ K (□), as a function of the mole fraction of benzene in the alkane-rich phase.

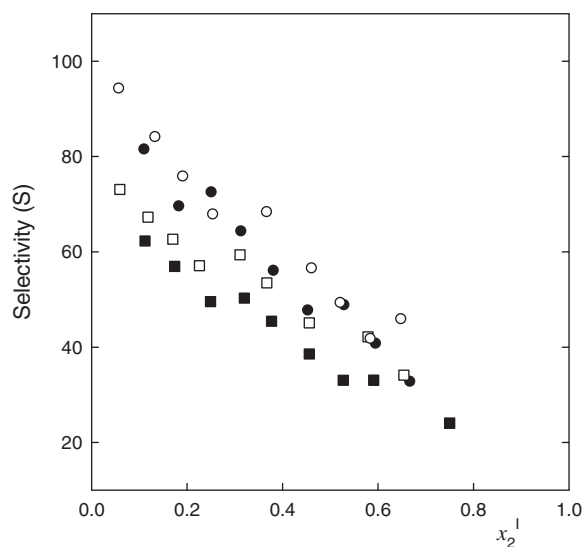


FIGURE 3. Selectivity for the ternary systems {hexane (1) + benzene (2) + [EPy][EtSO₄] (3)} at $T = 283.15$ K (●), {hexane (1) + benzene (2) + [EPy][EtSO₄] (3)} at $T = 298.15$ K (■), {heptane (1) + benzene (2) + [EPy][EtSO₄] (3)} at $T = 283.15$ K (○), and {heptane (1) + benzene (2) + [EPy][EtSO₄] (3)} at $T = 298.15$ K (□), as a function of the mole fraction of benzene in the alkane-rich phase.

5. Conclusions

The (liquid + liquid) equilibrium for the ternary systems of {hexane (1) + benzene (2) + 1-ethylpyridinium ethylsulfate (3)} and {heptane (1) + benzene (2) + 1-ethylpyridinium ethylsulfate (3)} were experimentally determined at $T = (283.15$ and $298.15)$ K and at atmospheric pressure, and the corresponding selectivity and distribution coefficient were calculated. The selectivity values are higher than unity for both studied systems at $T = (283.15$ and $298.15)$ K, and higher for the systems containing

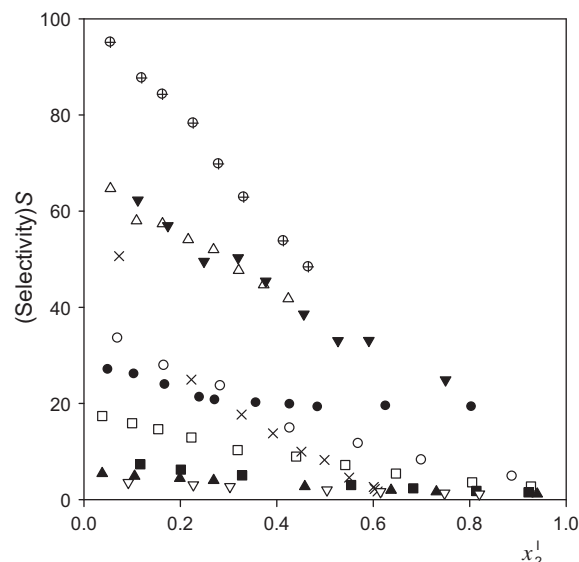


FIGURE 4. Selectivity of benzene for the ternary systems {hexane (1) + benzene (2) + solvent (3)} at $T = 298.15$ K: ▼, [EPy][EtSO₄] (this work), ●, [EMPy][EtSO₄] (from Ref. [37]), ○, [EMim][NTf₂] (from reference [17]), □, [BMim][NTf₂] (from reference [19]), ■, [OMim][NTf₂] (from reference [19]), ▲, [C₁₀Mim][NTf₂] (from reference [19]), ▽, [C₁₂Mim][NTf₂] (from reference [19]), ×, sulfolane (from reference [8]), ⊕, [EMim][ESO₄] (from reference [38]), Δ, [BMim][MSO₄] (from reference [39]).

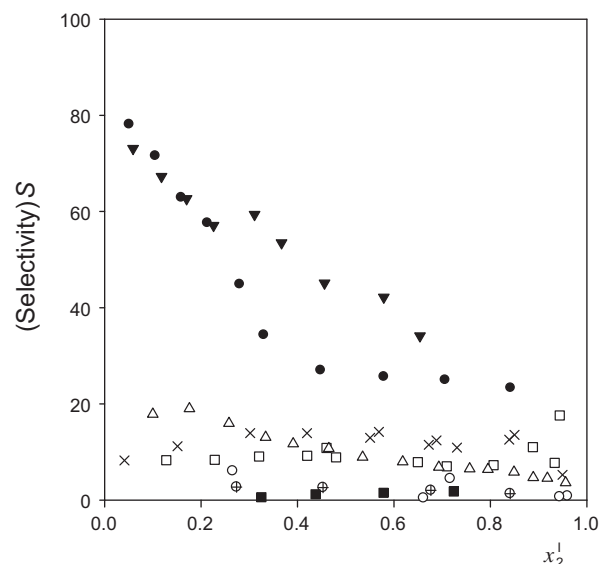


FIGURE 5. Selectivity of benzene for the ternary systems {heptane (1) + benzene (2) + solvent (3)} at $T = 298.15$ K: ▼, [EPy][EtSO₄] (this work), ●, [EMPy][EtSO₄] (from reference [37]), ○, [OMim][Cl] (from reference [36]), □, [HMim][BF₄] (from reference [40]), ×, [HMim][PF₆] (from reference [40]), ⊕, [EMim][OcsO₄] (from reference [41]), ■, [MOIM][MDEGSO₄] (from Ref. [41]), Δ, [BMpyr][NTf₂] (from reference [42]).

alkanes with larger chain, following the relation heptane > hexane. A small effect of temperature has been observed in the studied systems.

The consistency of the tie-lines was ascertained by applying the Othmer-Tobias and Hand equations, and the classical NRTL model was proven to acceptably correlate the experimental data of LLE. In general, the agreement between experimental and correlated tie-lines for the two systems investigated is remarkable.

TABLE 7

Othmer-Tobias and Hand equations parameters together with the correlation factor at every studied temperature for both ternary systems.

T/K	a	b	R ²
<i>Othmer-Tobias equation</i>			
{Hexane (1) + benzene (2) + [EPy][EtSO ₄] (3)}			
283.15	3.9815	1.6936	0.9855
298.15	3.9549	1.6595	0.9756
{Heptane (1) + benzene (2) + [EPy][EtSO ₄] (3)}			
283.15	3.3744	1.4908	0.9893
298.15	3.0404	1.3398	0.9861
	c	d	R ²
<i>Hand equation</i>			
{Hexane (1) + benzene (2) + [EPy][EtSO ₄] (3)}			
283.15	3.8139	1.6083	0.9829
298.15	3.7643	1.5605	0.9720
{Heptane (1) + benzene (2) + [EPy][EtSO ₄] (3)}			
283.15	3.1437	1.3843	0.9851
298.15	3.3246	1.4676	0.9910

TABLE 8

NRTL binary interaction parameters and deviations for LLE data of ternary systems with $\alpha = 0.1$.

T/K	$i-j$	$\Delta g_{ij}/(\text{kJ} \cdot \text{mol}^{-1})$	$\Delta g_{ji}/(\text{kJ} \cdot \text{mol}^{-1})$	σx	$\Delta \beta$
<i>{Hexane (1) + benzene (2) + [EPy][EtSO₄] (3)}</i>					
283.15	1–2	–3.980	7.149	0.131	1.655
	1–3	24.976	3.443		
	2–3	81.117	0.098		
298.15	1–2	–2.881	4.799	0.053	0.790
	1–3	17.229	3.772		
	2–3	91.005	0.451		
<i>{Heptane (1) + benzene (2) + [EPy][EtSO₄] (3)}</i>					
283.15	1–2	–5.592	7.076	0.045	0.469
	1–3	11.567	5.492		
	2–3	64.935	–2.887		
298.15	1–2	–3.832	5.984	0.069	0.841
	1–3	26.204	3.584		
	2–3	89.556	0.339		

Taken into account the results obtained in this work, the IL [EPy][EtSO₄] can be confirmed as a potential alternative solvent for the separation of aromatic and aliphatic hydrocarbons.

Acknowledgments

The authors are grateful to the Ministerio de Educación y Ciencia of Spain (project CTQ2007-61272), to the scholarships from Fundação para a Ciência e a Tecnologia (FCT, Portugal) (ref. SFRH/BDP/37775/2007 and ref. SFRH/BDP/48210/2008), and LSRE financing by FEDER/POCI/2010, for financial support.

References

- [1] S.H. Ali, H.M.S. Lababidi, S.Q. Merchant, M.A. Fahim, Fluid Phase Equilib. 214 (2003) 25–38.
- [2] B. Firnhaber, G. Emmerich, F. Ennenbach, U. Ranke, Erdoel Erdgas Kohle 116 (2000) 254–260.
- [3] T. Hambourger, L. Gouzien, P. Mikitenko, P. Bonfils, in: J.P. Wauquier (Ed.), Petroleum Refining, 2 Separation Processes, Editions Technip, Paris, 2000, pp. 359–456.
- [4] S.H. Hamid, M.A. Ali, Energy Sources 18 (1996) 65–84.
- [5] R.P. Tripathi, A.R. Ram, P.B. Rao, J. Chem. Eng. Data 20 (1975) 261–264.
- [6] B.S. Rawat, I.B. Gulati, J. Appl. Chem. Biotechnol. 26 (1976) 425–435.
- [7] R.M. de Fré, L.A. Verhoeve, J. Apple. Chem. Biotechnol. 26 (1976) 469–487.
- [8] J. Chen, L.P. Duan, J.G. Mi, W.Y. Feio, Z.C. Li, Fluid Phase Equilib. 173 (2000) 109–119.
- [9] J. Chen, Z.C. Li, L.P. Duan, J. Chem. Eng. Data 45 (2000) 689–692.
- [10] Y.J. Choi, K.W. Cho, B.W. Cho, Y.K. Yeo, Indian Eng. Chem. Res. 41 (2002) 5504–5509.
- [11] R. Krishna, A.N. Goswami, S.M. Nanoti, B.S. Rawat, M.K. Khana, J. Dobhal, Ind. J. Technol. 25 (1987) 602–606.
- [12] Y. Yorulmaz, F. Karpuzcu, Chem. Eng. Res. Des. 63 (1985) 184–190.
- [13] W. Wang, Z.M. Gou, S.L. Zhu, J. Chem. Eng. Data 43 (1998) 81–83.
- [14] J.A. Al-Sahhaf, E. Kapetanoviv, Fluid Phase Equilib. 118 (1996) 271–285.
- [15] M. Freemantle, Chem. Eng. News 76 (1998) 32.
- [16] U. Domanska, A. Pobudkowska, Z. Zolek-Tryznowska, J. Chem. Eng. Data 32 (2007) 2345–2349.
- [17] A. Arce, M.J. Earle, H. Rodríguez, K.R. Seddon, Green Chem. 9 (2007) 70–74.
- [18] U. Domanska, A. Pobudkowska, M. Królikowski, Fluid Phase Equilib. 259 (2007) 173–179.
- [19] A. Arce, M.J. Earle, H. Rodríguez, K.R. Seddon, J. Phys. Chem. B 111 (2007) 4732–4736.
- [20] S.I. Abu-Eishah, A.M. Dowaidar, J. Chem. Eng. Data 53 (2008) 1708–1712.
- [21] R. Wang, J. Wang, H. Meng, C. Li, Z. Wang, J. Chem. Eng. Data 53 (2008) 1159–1162.
- [22] A. Arce, M.J. Earle, H. Rodríguez, K.R. Seddon, A. Soto, Green Chem. 10 (2008) 1294–1300.
- [23] A. Arce, M.J. Earle, H. Rodríguez, K.R. Seddon, A. Soto, Green Chem. 11 (2009) 365–372.
- [24] A.B. Pereira, A. Rodríguez, J. Chem. Thermodyn. 41 (2009) 951–956.
- [25] E.J. González, N. Calvar, B. González, A. Domínguez, J. Chem. Eng. Data. doi:10.1021/jc900557u.
- [26] D.F. Othmer, P.E. Tobias, Ind. Eng. Chem. 34 (1942) 693–696.
- [27] D.B. Hand, J. Phys. Chem. 34 (1930) 1961–2000.
- [28] H. Renon, J.M. Prausnitz, AIChE J. 14 (1968) 135–144.
- [29] B. González, N. Calvar, E. Gómez, I. Domínguez, A. Domínguez, J. Chem. Eng. Data 54 (2009) 1353–1358.
- [30] J.A. Riddick, W.B. Bunger, T.K. Sakano, Organic Solvents, Wiley, New York, 1986.
- [31] Z.F. Wang, L.S. Wang, T.B. Fan, J. Chem. Eng. Data 52 (2007) 1866–1871.
- [32] T.M. Letcher, S. Wootton, B. Shuttleworth, C. Heyward, J. Chem. Thermodyn. 18 (1986) 1037–1042.
- [33] B. González, E.J. González, I. Domínguez, A. Domínguez, Phys. Chem. Liquids 55 (2010) 633–638.
- [34] G.W. Meindersma, A. Podt, A.B. de Haan, Fuel Process. Technol. 87 (2005) 59–70.
- [35] J. Zhu, J. Chen, C. Li, W. Fei, Sep. Purif. Technol. 56 (2007) 237–240.
- [36] T.M. Letcher, N. Deenadayalu, J. Chem. Thermodyn. 35 (2003) 67–76.
- [37] E.J. González, N. Calvar, B. González, A. Domínguez, J. Chem. Thermodyn. 41 (2009) 1215–1221.
- [38] J. García, A. Fernández, J.S. Torrecilla, M. Olet, F. Fernández, Fluid Phase Equilib. 282 (2009) 117–120.
- [39] J. García, A. Fernández, J.S. Torrecilla, M. Olet, F. Fernández, J. Chem. Eng. Data 55 (2010) 258–261.
- [40] T.M. Letcher, P. Reddy, J. Chem. Thermodyn. 37 (2005) 415–421.
- [41] N. Deenadayalu, K.C. Ngongo, T.M. Letcher, D. Ramjugernath, J. Chem. Eng. Data 51 (2006) 988–991.
- [42] A.B. Pereira, A. Rodríguez, AIChE J. 56 (2010) 381–386.